



BAULKHAM HILLS HIGH SCHOOL

2015

YEAR 11

TASK 2 JULY ASSESSMENT TASK

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black or blue pen
- Black pen is preferred
- Board-approved calculators may be used
- All relevant mathematical reasoning and/or calculations must be shown

Total marks – 45

(Pages 2-5)

Attempt Questions 1-4

Total Marks – 45

Attempt Questions 1 – 4

Answer each question on the appropriate page of your answer booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 1 (12 marks) Start on the appropriate page of your answer booklet

- a) A line makes an angle of 60° with the positive x -axis. What is the equation of the line if it passes through the point $(-1,4)$? 2

b) Evaluate:

(i) $\lim_{x \rightarrow 3} \frac{x^2 - 8x + 15}{x - 3}$ 2

(ii) $\lim_{x \rightarrow \infty} \frac{x^2 + 2}{2x^2 - 3x + 1}$ 2

c) Differentiate

(i) $\frac{1}{4x + 3}$ 2

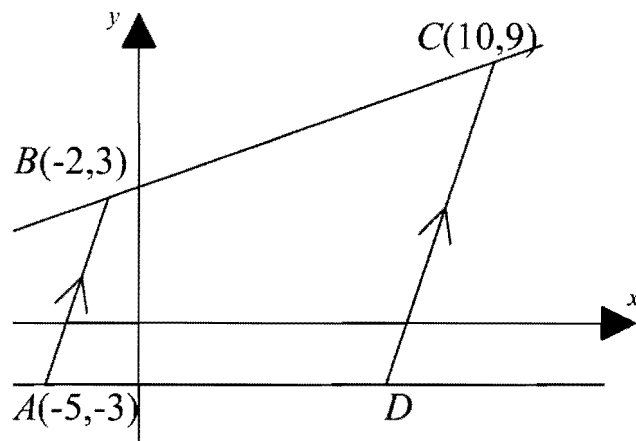
(ii) $\frac{2x}{x^2 + 3}$ 2

(iii) $\sqrt{2x^4 - 7x^3 + 1}$ 2

END OF QUESTION 1

Question 2 (12 marks) Start on the appropriate page of your answer booklet

- a) On a number plane the points $A(-5,-3)$, $B(-2,3)$, $C(10,9)$ and D form a trapezium in which AD is parallel to the x axis and $AB \parallel CD$.



- (i) Show that the equation of the line DC is $2x - y - 11 = 0$ 2
 - (ii) Find the coordinates of D . 1
 - (iii) Find the perpendicular distance from B to DC 2
 - (iv) Find the coordinates of the midpoint of BD . 1
 - (v) The point E lies on the line BA produced such that $BCDE$ is a rhombus. Find the coordinates of E . 1
- b) Joel is trying to find the equation of a line perpendicular to $x - 2y + 13 = 0$ which passes through the point of intersection of the lines $3x + 2y - 1 = 0$ and $7x - 11y + 3 = 0$.

As part of his working Joel has correctly set up the following equation:

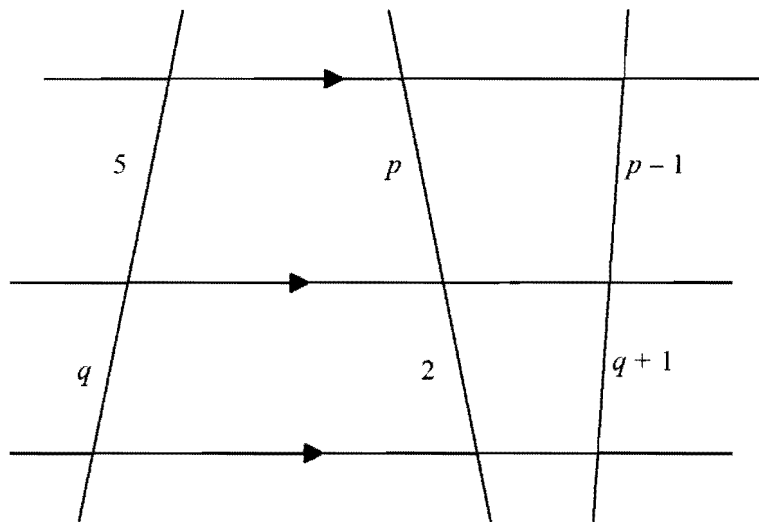
$$3x + 2y - 1 + k(7x - 11y + 3) = 0$$

- (i) Show that the gradient of the line is $\frac{3 + 7k}{11k - 2}$ 2
- (ii) Find the value of k 2
- (iii) Hence determine the required equation of the line. 1

END OF QUESTION 2

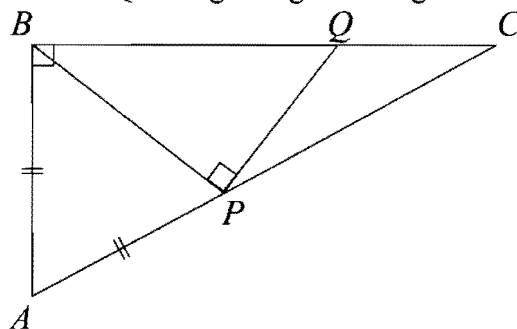
Question 3 (12 marks) Start on the appropriate page of your answer booklet

- a) The diagram below comprises three parallel lines.



- i) Explain why $\frac{5}{q} = \frac{p}{2}$ 1
- ii) Find the exact value of p and q 2
- b) Find the equation of the tangent to the curve $y = 3x^3 - 4x^2 + x - 1$ at the point where $x = 1$. 3

- c) Triangles ABC and BPQ are right angled triangles and $AB=AP$



Copy the diagram into your answer booklet

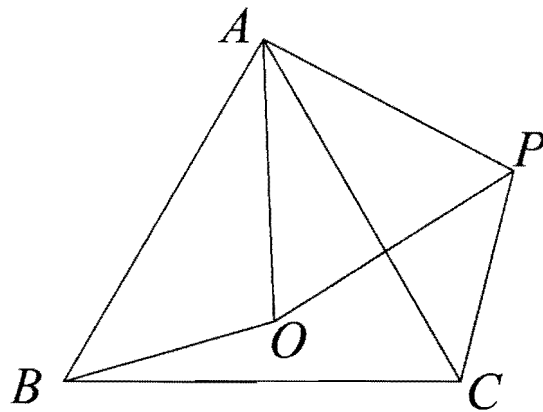
- (i) Prove $\angle CBP = \angle CPQ$ 3
- (ii) Hence prove that $PC^2 = BC \times QC$ 3

END OF QUESTION 3

Question 4 (9 marks) Start on the appropriate page of your answer booklet

- a) The parabola $y = ax^2 + bx + c$ passes through the point $(0, -2)$. At the point $(2, -3)$ on this curve, the equation of the tangent is $2x - y - 7 = 0$.
Find the values of a , b and c .

- b) In the figure triangles ACB and APC are equilateral.



NOT TO SCALE

Copy this diagram into your answer booklet.

- | | | |
|-------|--|----------|
| (i) | Prove that $\angle BAO = \angle PAC$ | 2 |
| (ii) | Prove $\triangle AOB \equiv \triangle APC$ | 3 |
| (iii) | Hence, prove $OB = CP$. | 1 |

END OF EXAMINATION

YR 11 JUNE 2015 2 UNIT

Q1

a) $m = \tan 60^\circ$
 $= \sqrt{3}$ ✓

$$y - y_1 = m(x - x_1)$$

$$y - 4 = \sqrt{3}(x + 1)$$

$$y = \sqrt{3}x + \sqrt{3} + 4$$
 ✓
 (OR $\sqrt{3}x - y + \sqrt{3} + 4 = 0$)

- (2) correct solution
 (1) finding gradient
 or using incorrect gradient
 to correctly find line

b) i) $\lim_{x \rightarrow 3} \frac{x^2 - 8x + 15}{x - 3}$
 $= \lim_{x \rightarrow 3} \frac{(x - 5)(x - 3)}{x - 3}$
 $= \lim_{x \rightarrow 3} (x - 5)$ ✓
 $= 3 - 5$
 $= -2$ ✓

- (2) correct solution
 (1) incorrect factorisation
 (1) and subsequent cancelling

ii) $\lim_{x \rightarrow \infty} \frac{x^2 + 2}{2x^2 - 3x + 1}$
 $= \lim_{x \rightarrow \infty} \frac{\frac{x^2}{x^2} + \frac{2}{x^2}}{\frac{2x^2}{x^2} - \frac{3x}{x^2} + \frac{1}{x^2}}$ ✓
 $= \lim_{x \rightarrow \infty} \frac{1 + \frac{2}{x^2}}{2 - \frac{3}{x} + \frac{1}{x^2}}$
 $= \frac{1 + 0}{2 - 0 + 0}$
 $= \frac{1}{2}$ ✓

- (2) correct solution
 with/without working
 (1) attempts to use $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$
 or (1) divides throughout by x^2

$$1c \ i) \ \frac{d}{dx} (4x+3)^{-1} = -1 (4x+3)^{-2} \cdot 4 \checkmark$$

$$= \frac{-4}{(4x+3)^2} \checkmark$$

(2) correct solution

(1) differentiation without multiplying by 4.

$$ii) \ \frac{d}{dx} \left(\frac{2x}{x^2+3} \right) = \frac{(x^2+3) \cdot 2 - 2x \cdot 2x}{(x^2+3)^2} \checkmark$$

$$= \frac{2x^2+6-4x^2}{(x^2+3)^2}$$

(2) correct solution

(1) attempts to use quotient rule

$$= \frac{6-2x^2}{(x^2+3)^2} \text{ or } \frac{2(3-x^2)}{(x^2+3)^2} \checkmark$$

$$(iii) \ \frac{d}{dx} (2x^4-7x^3+1)^{\frac{1}{2}} = \frac{1}{2} (2x^4-7x^3+1)^{-\frac{1}{2}} \cdot (8x^3-21x^2) \checkmark$$

$$= \frac{8x^3-21x^2}{2\sqrt{2x^4-7x^3+1}} \text{ or } \frac{x^2(8x-21)}{2\sqrt{2x^4-7x^3+1}} \checkmark$$

(2) correct soln

(1) differentiates without using chain rule

$$2 \ a) \ i) \ m_{AB} = m_{AB}$$

$$= \frac{3 - (-3)}{-2 - (-5)}$$

$$= \frac{6}{3}$$

$$= 2 \checkmark$$

(2) correct solution

(1) correct gradient or finds equation of line using incorrect gradient

$$\text{Using } y - y_1 = m(x - x_1)$$

$$y - 9 = 2(x - 6)$$

$$y = 2x - 12 + 9$$

$$y = 2x - 3 \text{ or } 2x - y - 3 = 0 \checkmark$$

$$ii) \ \text{when } y = -3$$

$$2x + 3 - 11 = 0$$

$$2x = 8$$

$$x = 4$$

$$\therefore D \text{ is } (4, -3) \checkmark$$

(1) correct solution

2a (iii) Perp Dist: $\left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$

$= \left| \frac{2x - 2 - 3 - 11}{\sqrt{2^2 + 1^2}} \right|$ ✓

$= \left| \frac{-18}{\sqrt{5}} \right|$

$= \frac{18\sqrt{5}}{5}$ ✓

② correct solution
① substitutes into perpendicular distance formula.

(iv) Midpoint = $\left(\frac{-2+4}{2}, \frac{3+(-3)}{2} \right)$

$= (1, 0)$ ✓

① correct solution

(v) $\begin{cases} 1 = \frac{x+10}{2} \\ 0 = \frac{y+9}{2} \end{cases}$

① correct solution

$2 = x+10$

$x = -8$

$y+9=0$

$\therefore \text{E is } (-8, -9)$ ✓

2b (i) $3x+2y-1 + 7kx - 11ky + 3k = 0$

$(7k+3)x + 3k-1 = (11k-2)y$ ✓

$\therefore m = \frac{7k+3}{11k-2}$ ✓

② correct solution
① expresses in general form or $y=mx+b$

(ii) grad $\perp = -2 \therefore -2 = \frac{7k+3}{11k-2}$ ✓

$-2(11k-2) = 7k+3$

$-22k+4 = 7k+3$

$-29k = -1$

$k = \frac{1}{29}$ ✓

$4y = 11x+3$
 $y = \frac{11}{4}x + \frac{3}{4}$ ② correct solution
 $m_{\perp} = -2$ ① equating gradient with -2

(iii) $3x+2y-1 + \frac{1}{29}(7x-11y+3) = 0$

$87x+58y-29+7x-11y+3=0$

$94x+47y-26=0$ ✓

① correct solution

3a) (i) $\frac{5}{q} = \frac{p}{2}$ (ratio of intercepts on parallel lines) ① correct reason

(ii) $\frac{5}{q} = \frac{p}{2}$
 $10 = pq$

$\frac{p}{2} = \frac{p-1}{q+1}$
 $pq + p = 2p - 2$ ②
 $10 + p = 2p - 2$
 $12 = p$
 $\therefore q = \frac{10}{p}$
 $= \frac{10}{12}$
 $q = \frac{5}{6}$

② correct solution
 ① uses ratio of intercept theorem to set up simultaneous equations and attempts to solve

b) $\frac{dy}{dx} = 9x^2 - 8x + 1$

when $x=1$, $\frac{dy}{dx} = 9 - 8 + 1$
 $= 2$
 $f(1) = 3 - 4 + 1 = 0$

$y + 1 = 2(x - 1)$
 $y + 1 = 2x - 2$
 $y = 2x - 3$ or $2x - y - 3 = 0$ ✓

③ correct solution
 ② either gradient or y-value or equation contains error
 ① finds gradient or y-value or equation.

c) i) let $\angle PBQ = x$
 $\angle PBA = 90^\circ - x$ (adjacent complementary $\angle$'s) ✓
 $\angle BPA = 90^\circ - x$ (equal $\angle$'s opposite equal sides in $\triangle PAB$)
 $AB = AP$ given ✓
 $\angle APB + \angle BPQ + \angle QPC = 180^\circ$ (angle sum of straight line) ✓
 $90^\circ - x + 90^\circ + \angle QPC = 180^\circ$
 $\angle QPC = 180^\circ - 90^\circ + x = 90^\circ + x$

③ correct solution
 ② correct with at least 1 reason
 ① correct with no reasons or expression for $\angle BPA$ with reason
 $\angle PBQ$ as req'd. ✓

C (iii) In $\triangle BCP$ and $\triangle PCQ$

LC is common

$$\angle CBP = \angle PCQ \quad (\text{proven in (ii)})$$

$\therefore \triangle BCP \parallel \triangle PCQ$ (two pairs of matching angles equal) ✓

$$\frac{BC}{PC} = \frac{PC}{QC} \quad (\text{matching sides of similar } \triangle BCP \text{ \& } \triangle PCQ \text{ in same ratio}) \quad \checkmark$$

$$\therefore PC^2 = BC \cdot QC$$

- ③ correct solution
- ② correct with reason
- ① correct with no reasons or proves similar without reasons

Q4

a)

$$y = ax^2 + bx + c$$

since curve passes through $(0, -2)$

$$-2 = 0 + 0 + c$$

$$\therefore y = ax^2 + bx - 2$$

since $(2, -1)$ lies on curve

$$-1 = 4a + 2b - 2$$

$$-1 = 4a + 2b$$

$$4a + 2b = -1$$

$$\frac{dy}{dx} = 2ax + b$$

When $x = 2$, $\frac{dy}{dx} = 2$ gradient of tangent

$$\therefore 4a + b = 2$$

- ③ correct
- ② forms simultaneous equations and attempts to solve (with c)
- ① finds c or $4a + 2b = -1$ or $4a + b = 2$

①-②

$$b = -3$$

sub in ① $4a - 3 = 2$

$$4a = 5$$

$$a = \frac{5}{4}$$

$$\therefore y = \frac{5}{4}x^2 - 3x - 2$$

4 b(i)

Let $\angle OAC = x$

$\angle BAC = \angle OAP = 60^\circ$ (angles in equilateral Δ 's)

$\angle BAO = 60^\circ - x$

$\angle PAC = 60^\circ - x$

$= \angle BAO$

$\therefore \angle BAO = \angle PAC$

Correct solution

(1) Attempt to use 60° angle to find an expression

(ii) In ΔAOB and ΔAPC

$AB = AC$ (equal sides of equilateral ΔABC) ✓

$OA = PA$ (equal sides of equilateral ΔOAP) ✓

$\angle BAO = \angle PAC$ (proven in (i)) ✓

$\therefore \Delta AOB \cong \Delta APC$ (SAS)

(iii) $OB = CP$ (matching sides in congruent Δ 's AOB APC) ✓